

AN EFFECTIVE SOLUTION FOR TIME CRUNCHED CYCLISTS?

knowledgeiswatt.substack.com/p/44-block-hit-periodization-an-effective

Knowledgeiswatt

November 20, 2024



Photo by [Markus Winkler](#) on [Unsplash](#)

When considering the annual training plan, linear classic periodization (a progressive increase in volume first and intensity then while keeping or slightly reducing the volume) is the most used strategy by successful endurance cyclists. However, for time-crunched cyclists this solution is not always feasible. In fact, they can usually train only ~1 hour from Monday to Friday, with the possibility to go for long rides only on some weekends. So, they lack time to progressively increase volume and build the linear classic periodization. As a consequence, many usually insert a relevant amount of high intensity training early in the periodization. In this way, they soon reach a plateau in volume and intensity that is usually kept for several months without progressive overloading the body.

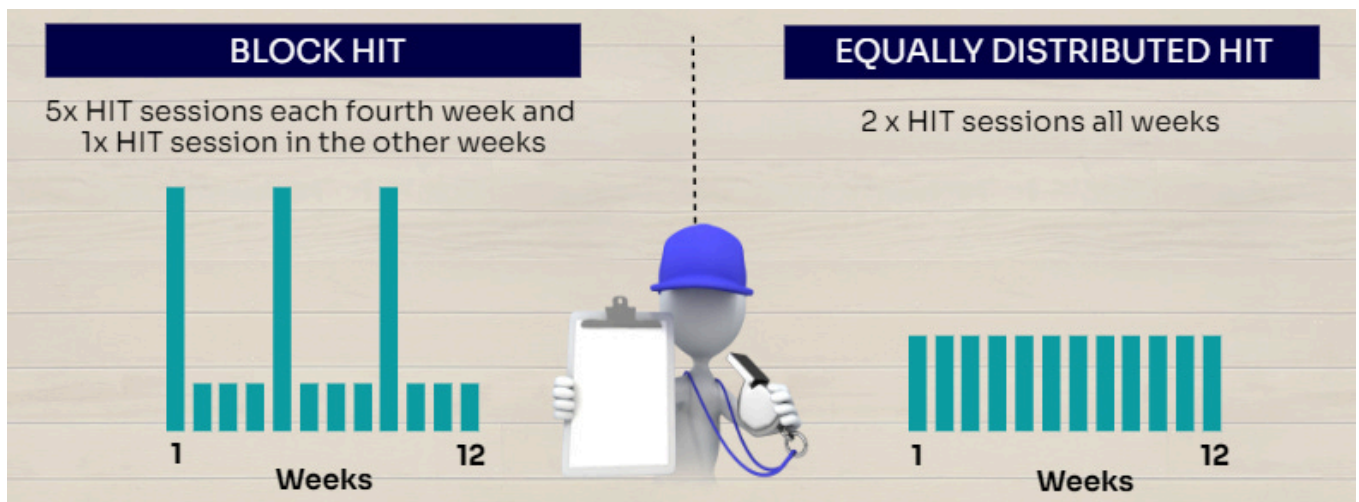
For these cyclists, a good alternative approach could be **high-intensity (HIT) block periodization**.

Can HIT block periodization be an effective alternative periodization to optimize cycling performance?

A study published in 2014 by Prof Bent Rønnestad and Colleagues (Lillehammer University, Norway) on *Scandinavian Journal of Medicine & Science in Sports* tried to answer this question. (1)

WHAT DID THEY DO?

- They recruited 18 male well-trained cyclists (VO₂max 63, age 33) which for 12 weeks were divided into two different groups/periodization: **block HIT vs HIT distributed equally**.
- For the 12 training weeks, the two groups completed the same volume of low intensity (-8/10 hours per week) and the same number of high intensity training session (24, consisting in 5x6min or 6x5min at maximum sustainable efforts with 2.5 recovery in between) with the only between groups difference consisting in how they periodized the high intensity sessions:
 - **block HIT: they performed 5x week HIT sessions on week 1, 5 and 9 while only 1 HIT session during the other weeks.**
 - **HIT distributed equally: they performed 2x week HIT sessions throughout all the period.**

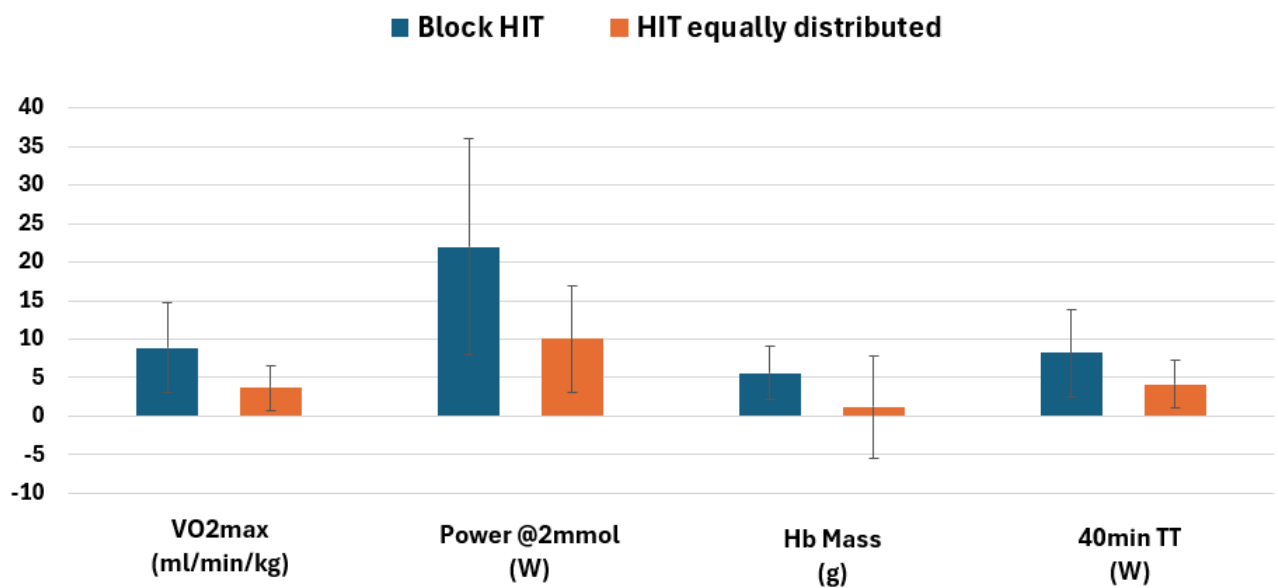


KisW representation of the study design. (1)

WHAT DID THEY FIND?

Block HIT Periodization tended to induce superior aerobic performance and physiological adaptations compared to HIT equally distributed:

- VO₂max: $+8.8 \pm 5.9\%$ vs $+3.7 \pm 2.9\%$
- Power output at 2 mmol/L of blood lactate: $+22 \pm 14\%$ vs $+10 \pm 7\%$
- Hbmass (haemoglobin mass): $+5.6 \pm 3.5\%$ vs $+1.2 \pm 6.6\%$,
- 40 min Time Trial (TT) performance: $+8.2 \pm 5.7\%$ vs $+4.1 \pm 3.1\%$



% changes after the 12 weeks in the two different groups. Data from Rønnestad et al. (1)

Note that the large % improvements in both groups (especially in power @2mmol) can be related to the fact that all the participants did not perform any high intensity training in the weeks before entering the study.

PRACTICAL TAKEAWAYS:

12 weeks of HIT block periodization provides superior performance and physiological adaptations compared to an equally distributed HIT periodization when training volume is kept constant at around 8/10 hrs per week.

GABRIELE'S COMMENT

How can we apply these data?

- As said above, I see this HIT block periodization as a good strategy for time- crunched cyclist, as they can't follow the classic linear periodization without maintaining very similar volume and intensity for several weeks, which could lead to staleness in their performance level.
- These data can be also applied to professional/no time crunched cyclists. After having already increased first volume and then intensity through the periodization, HIT block periodization could be used at the end of periodization (1 or two blocks), just before the goal race/s. This is very similar to what riders contending grand tours usually do when entering one-week stage races (which represent a huge increase in high intensity volume) after the winter preparatory period.
- As HIT training put a high level of strain on the body, before entering a HIT block, make sure to be fresh and rested so that the body can 'absorb' the block.

Thanks for reading! Share with your team!

Gabriele Gallo, PhD in Exercise and Sport Sciences

REFERENCES:

-
1. Rønnestad BR, Ellefsen S, Nygaard H, Zacharoff EE, Vikmoen O, Hansen J, Hallén J. Effects of 12 weeks of block periodization on performance and performance indices in well-trained cyclists. Scand J Med Sci Sports. 2014 Apr;24(2):327-35.



[21 Likes](#)

[2 Restacks](#)